

TURBULENT PIPE FLOW ANALYZER AND ENERGY SPECTRUM VISUALIZER

An Interactive Engineering Tool for Turbulence Analysis

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➤ Objective

This project aims to create an interactive tool to analyze turbulent pipe flows and visualize turbulence energy distribution, combining engineering calculations with dynamic visualizations.

➤ Background

Turbulence is a complex fluid flow regime characterized by chaotic, stochastic property changes. It plays a major role in aerospace, mechanical, and energy systems. Predicting flow behavior, pressure losses, and energy cascades is essential in engineering design.

➤ Methods

The analysis is based on the core principles of fluid mechanics:

- Reynolds number calculation to classify flow regimes (laminar, transitional, turbulent)
- Friction factor determination using exact and empirical formulas (Swamee-Jain approximation for turbulence)
- Pressure drop estimation via Darcy-Weisbach equation
- Turbulent intensity and turbulent kinetic energy estimation for deeper characterization

The energy cascade is simulated following Kolmogorov's $-5/3$ law, representing how energy transfers across scales in turbulent flows.

➤ Technologies Used

- **Microsoft Excel:** Used for engineering calculations, dynamic dashboards, live-updating graphs, conditional formatting, and interactive controls (sliders).
- **MATLAB:** Used to generate a log-log plot of the turbulence energy spectrum based on Kolmogorov's theory.

➤ Results and Features

- Fully automated engineering calculator for turbulent flow analysis
- Live updating graphs for Pressure Drop vs Flow Rate, and Friction Factor vs Reynolds Number
- Conditional formatting for flow regime visualization (green/yellow/red)
- Interactive slider to control Flow Rate dynamically
- MATLAB visualization of the turbulence energy cascade

➤ Bonus Features

- Professional layout and styling
- Automatic linking between inputs, calculations, and graphs
- MATLAB-based energy cascade plotting with labeled $-5/3$ slope

➤ Future Work

- Extend the tool to non-circular ducts and complex geometries
- Add CFD (Computational Fluid Dynamics) simulation data comparison
- Integrate machine learning to predict flow regime transitions